MOHAMMAD NAVED

AI-Assisted Developer | Prompt Engineer | Full-Stack Product Builder

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PROFESSIONAL SUMMARY

AI-assisted developer who designs, ships, and tests full-stack web platforms, commercial IoT systems, and LLM-powered applications by directing autonomous AI coding agents and workflows. Hands-on experience integrating multi-model LLM APIs (Claude, Gemini, NVIDIA NIM, GLM), architecting structured prompt pipelines, natural-language-to-SQL querying, and cloud deployment. Owns end-to-end system architecture, interface specifications, testing gates, and edge-case debugging; directs AI agents for rapid, clean code execution. Shipped an officially certified Google Smart Home system (71 of 75 automated test cases passed), a multi-tenant cloud pharmacy SaaS, and an AI analytics platform. B.Tech in Computer Science.

TECHNICAL SKILLS

AI & LLM Engineering: Multi-LLM APIs (Claude, Gemini, NVIDIA NIM, GLM), Prompt Engineering, Agentic Workflows, Natural-Language-to-SQL, AI Coding Agents & Autonomous Workflows

Web Technologies: React 19, Next.js 15/16, TypeScript, JavaScript, Tailwind CSS v4, Vite, HTML5/CSS3

Backend & Databases: FastAPI (Python), Node.js, REST APIs, Supabase, PostgreSQL, SQL, DuckDB, Python, Power BI

IoT, Cloud & DevOps: ESP32 Firmware, MQTT Protocol, AWS IoT Core, AWS App Runner, Docker, Vercel, GitHub Actions CI/CD

PROFESSIONAL EXPERIENCE

Codtrex AI — Boutique AI Engineering Studio

2025 - Present

Founder and AI Engineer • codtrex.vercel.app • React 19, Vite, Tailwind CSS, Supabase, Vercel

- Run a boutique AI engineering studio delivering production-grade web applications, multi-LLM integrations, and IoT platforms with 14-20 day rapid turnaround.
- Architected multi-provider AI failover gateway (Claude, Gemini, NVIDIA NIM, GLM, Groq) with context caching and dynamic language detection.
- Built and deployed the studio website with an interactive AI chat assistant, productized service tiers, and architecture case studies.
- Deliver client products by defining data schemas, writing structured prompts for AI coding agents, reviewing code diffs, and running regression tests before deployment.

FEATURED PROJECTS

4Layers — Smart Home IoT Ecosystem (Google Certified)

2025 - Present

ESP32 C++, AWS IoT Core, MQTT, React Native, FastAPI, Docker, Google Smart Home API

- Passed 71 of 75 automated test cases in the official Google Home Test Suite; devices verified, listed, and fully functional in the Google Home app.
- Designed 3-source real-time state synchronization (ESP32 hardware, AWS IoT Core MQTT, Google Home) ensuring physical switches, mobile app, and voice commands stay consistent.
- Diagnosed and resolved a certification-blocking issue where Google Home reverted local device states during async verification, by engineering a custom QA Hold Mode with automated TTL.
- Implemented over-the-air (OTA) firmware updates with version pinning and dual-partition (A/B) rollback safety for zero-touch device maintenance.
- Designed custom ESP32 hardware: 4-channel relay switch boards and multi-speed fan regulators; provisioned per-device mTLS certificates through AWS IoT Core.

MediStock — Cloud Pharmacy Management SaaS & POS

2025 - Present

React 18, Supabase, PostgreSQL, Capacitor Android, Vercel • medistock-pos.vercel.app

- Shipped a B2B cloud pharmacy platform covering inventory management, FEFO automated batch allocation, expiry tracking, billing, and supplier accounting.
- Designed a multi-tenant PostgreSQL schema with Row Level Security (RLS) to enforce strict store-level data isolation across accounts.

PROJECTS (CONTINUED)

DataLens AI — Natural-Language-to-SQL Analytics Platform *2025*

React, TypeScript, Recharts, Gemini API, DuckDB, SQL

- Built an AI analytics platform with a natural-language-to-SQL chat interface enabling non-technical users to query CSV and Excel datasets through plain English questions.
- Engineered deterministic SQL execution safeguards: enforced read-only database roles and executed queries in-memory using DuckDB to prevent dangerous operations.
- Developed ML Lab module for model output explanation and business insight surfacing using Gemini-powered prompt templates and schema injection.
- Designed the end-to-end data pipeline, SQL execution engine, and dynamic interactive chart rendering using Recharts.

UsedTech Market — C2C Electronics Marketplace *2025*

Next.js 15/16, React 19, TypeScript, Supabase, Tailwind CSS v4

- Designed and shipped a consumer-to-consumer marketplace for buying and selling certified used electronics with real-time listings and seller dashboards.
- Built on Next.js App Router with server-side rendering for SEO-optimized product pages and Supabase real-time subscriptions for instant listing updates.
- Defined the full relational data model, authentication flow, image upload pipelines, and complete listing lifecycle (draft, published, sold).
- Implemented role-based access control (RBAC) and seller verification checks to safeguard marketplace transactions against fraud.

Interactive Portfolio & Automation Pipeline *2025*

React 19, TypeScript, Tailwind CSS v4, Node.js, Puppeteer, Gemini API • mohammad-naved-portfolio.vercel.app

- Built a responsive, dark/light mode single-page portfolio featuring an interactive Gemini-powered AI chatbot assistant and terminal emulator.
- Automated resume PDF generation from HTML source using a Node.js and Puppeteer pipeline for pixel-perfect A4 rendering.
- Integrated hardware-accelerated micro-interactions, dark/light theme persistence, and automated Vercel CI/CD deployment pipelines.

AI ENGINEERING & VERIFICATION METHODOLOGY

- **Contract & Schema First:** Define strict data contracts, TypeScript interfaces, and PostgreSQL schemas before prompting AI agents, eliminating API hallucinations and schema drift.
- **Automated Verification Gates:** Direct AI agents to write unit and integration test suites; execute automated test suites (e.g., Google Home 75-test certification suite) before code merge.
- **Deterministic Failsafes:** Enforce read-only locks, database row-level security (RLS), and in-memory execution boundaries (DuckDB) for high-risk operations.
- **Network & Payload Inspection:** Profile API response latencies, inspect browser network payloads, and analyze PostgreSQL query execution plans rather than relying solely on visual outputs.

EDUCATION

B.Tech, Computer Science — RGPV University, Indore *Graduated: 2025*

Rajiv Gandhi Proudyogiki Vishwavidyalaya • Madhya Pradesh, India

- **Academic Focus:** Systems Architecture, Full-Stack Web Development, Database Engineering, and Distributed Systems.
- **Relevant Coursework:** Data Structures & Algorithms, Database Management Systems (DBMS), Operating Systems, Computer Networks, Object-Oriented Programming (OOP), Software Engineering.

CERTIFICATIONS & ACHIEVEMENTS

- **Google Smart Home Certification:** 4Layers IoT Ecosystem (71 of 75 automated test cases passed in official Google Home Test Suite).
- **Generative AI with Gemini:** Prompt Engineering and LLM Application Development.
- **Python for Data Analysis and Visualization:** Exploratory data analysis, Pandas, and automated visualization pipelines.
- **SQL for Data Analytics:** Multi-table relational queries, joins, subqueries, and database schema normalization.
- **Power BI:** Interactive dashboard design, DAX data modeling, and business intelligence reporting.